FINANCIAL ECONOMETRICS

MODULE 2 | LESSON 1

OLS REGRESSION ASSUMPTIONS AND WEIGHTED LEAST SQUARE REGRESSION

Reading Time	60 minutes
Prior Knowledge	OLS regression analysis, Linear Algebra
Keywords	Homoscedasticity, Heteroskedasticity, Independent and identically distributed, Weighted least square regression, Breusch-Pagan Test

In the last module, we discussed the applications of linear regression models and some of their assumptions. We learned how to handle multicollinearity when independent variables are highly correlated. We also went through methods to transform variables when they don't meet the assumptions in a regression model. In this module, we will dive deeper into and offer variations of regression analysis. We will first review main assumptions for an ordinary least square model. Then, we will focus on the heteroskedasticity of a regression model and introduce the weighted least square model as a way to handle the issue.

We are also introducing a new working example in this module. We will analyze some factors that might impact the foreign exchange markets. We will use different regression methods we learn in each lesson to conduct the analysis.

In [1]:
```
# Libraries
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import statsmodels.api as sm
import statsmodels.formula.api as smf
import statsmodels.stats.api as sms

plt.rcParams["figure.figsize"] = (14, 8)  # Figure size and width
```

In [2]:
```
# Import data
data_set = pd.read_csv("M2. module_2_data.csv")
```

1. Assumptions of Ordinary Least Square Regression

Ordinary least square (OLS) regression models are based on the following assumptions.

1.1 Linear Relationship

The OLS regression model assumes that the dependent variable has a linear relationship with independent variables. If Y is the dependent variable and X_1, X_2 and X_3 are independent variables and e is the error term, we can write an OLS regression model as follows:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + e$$

From Module 1, we learned how to use variable transformation methods to ensure this relationship exists.

1.2 No Multicollinearity

This assumption states that independent variables should be linearly independent to each other. In reality, many independent variables are correlated to each other. Some might have low correlations and some might have high correlations. We learned from the last module that if independent variables are highly correlated, we can check their correlations or VIFs and drop variables. We can also apply principal component analysis or factor analysis to address the issue.

1.3 Independent Error Terms

This assumption for OLS means that the error terms from each observation will be independent of each other.

Assume we have these two observations from the dataset $(y_1, X_{11}, X_{21}, X_{31})$ and $(y_2, X_{12}, X_{22}, X_{32})$. We can plug these two observations into an OLS regression model system.

$$y_1 = \beta_0 + \beta_1 X_{11} + \beta_2 X_{21} + \beta_3 X_{31} + e_1$$
$$y_2 = \beta_0 + \beta_1 X_{12} + \beta_2 X_{22} + \beta_3 X_{32} + e_2$$

This assumption states that e_1 and e_2 are independent of each other. If they are not independent, we call them **autocorrelated**.

1.4 Error Terms are Normally Distributed

From our example, this means that e_1 and e_2 have to be normally distributed.

1.5 Expected Values of Error Terms are 0

From our example, the expected values of e_1 and e_2 have to be 0.

1.6 Error Terms Have the Same Variance (Homoscedasticity)

When the variances of error terms are constant, we call the error terms **homoscedastic.** If the error terms are not constant, we call them **heteroskedastic.**

With all the above assumptions about error terms, we usually call the error terms **IID: independent and identically distributed**. With the assumptions of 1.4, 1.5, and 1.6, we can summarize the error terms as follows:

$$e \sim N(0, \sigma^2)$$

Now we can write the multiple regression mean function as follows:

$$E(Y|X) = \beta X$$

And variances of the errors are constant.

$$\mathrm{Var}(Y|X) = \mathrm{Var}(e|X) = \delta^2$$

Because the variance function is for multiple regression, $\mathrm{Var}(e)$ is actually a covariance matrix.

$$\mathrm{Var}(e) = \begin{pmatrix} \delta^2 & 0 & 0 & \cdots & 0 \\ 0 & \delta^2 & 0 & \cdots & 0 \\ 0 & 0 & \delta^2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \delta^2 \end{pmatrix}$$

We can see the values on the diagonal of the matrix are the constant variance, and the values off the diagonal of the matrix are 0. Because the assumption error terms are independent, the covariances of different error terms are 0.

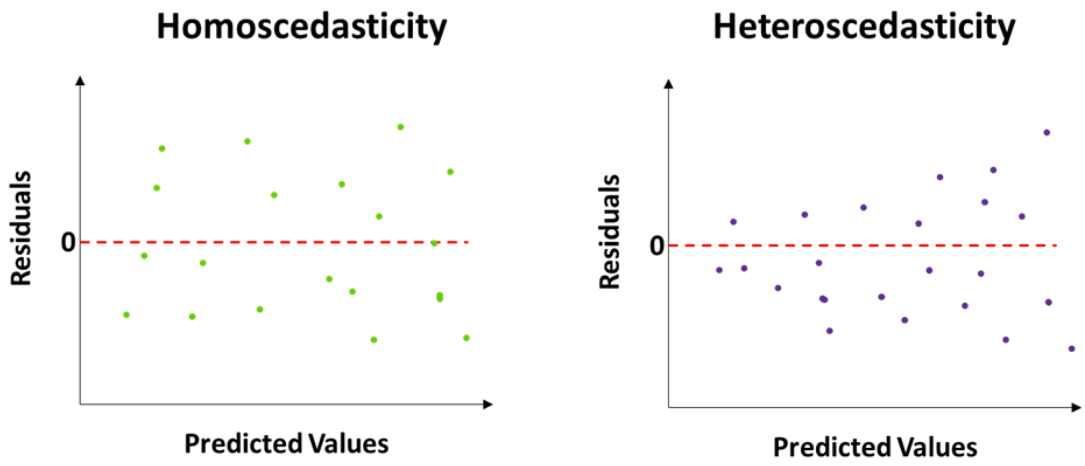
Now we want to see what happens when any of the above assumptions do not exist. In the last module, we talked about how to handle the situations when the linear relationship and low multicollinearity assumptions are violated. In the following section, we will turn our focus on the assumption of homoscedasticity. We want to see how to detect when the homoscedasticity assumption does not hold up and what method we can use to run a regression analysis when heteroskedasticity exists. Particularly, we will introduce a weighted least square regression to handle heteroskedasticity.

2. Weighted Least Square Regression

2.1 How to Detect Heteroskedasticity

As we mentioned in the last section, heteroskedasticity occurs when error terms in a regression analysis are not constant among observations. After running an OLS regression model, we can draw a scatter plot between predicted values vs. residuals to check for homoscedasticity (or heteroskedasticity). **Residuals** are the differences between measured values of the dependent variables and predicted values of dependent variables from the OLS regression model.

Figure 1: Homoscedasticity vs. Heteroskedasticity



From Figure 1, we can see the residuals scatter around the 0 line evenly in the homoscedasticity graph on the left. It means the variance for residuals is very constant along the 0 line. However, in the heteroskedasticity graph on the right, the residuals are more concentrated when the predicted values are small, and the residuals are more spread out when the predicted values are large. The data shows a cone or a megaphone pattern in the graph. It implies that as the predicted values increase, the variances increase too. The variances are not constant.

Graphs are visual aids. Sometimes, the graphs can make it difficult to spot the issue of heteroskedasticity. There are several formal methods to detect heteroskedasticity. The **Breusch-Pagan (BP) test** is a formal statistical test to detect the issue.The idea of the BP test is that it assumes that the variance and independent variables have a linear relationship. The test then runs a regression of the squared residuals from the original regression on the independent variables. Then, a BP test checks the following hypothesis:

$$H_0 : \text{Homoscedasticity exists}$$

$$H_1 : \text{Heteroskedasticity exists}$$

This is a *Chi-square* test. If the p-value is less than the significant level of 0.05, we reject the null hypothesis. The residuals are not constant, so there is a heteroskedasticity issue.

There are other tests to check for heteroskedasticity. One of them is the **White test,** which tests the same hypothesis as the BP test. However, the White test does not assume a linear relationship between the variance and independent variables.

2.2 Basics of Weighted Least Square Regression

When the error term variances are not constant, the covariance matrix for n observations will be like the following:

$$\mathrm{Var}(e) = \begin{pmatrix} \delta_1^2 & 0 & 0 & \cdots & 0 \\ 0 & \delta_2^2 & 0 & \cdots & 0 \\ 0 & 0 & \delta_3^2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \delta_n^2 \end{pmatrix}$$

where $\delta_1^2 \neq \delta_2^2 \neq \delta_3^2 \neq \cdots \neq \delta_n^2$

Let's assume the variance of the error term of observation i is as follows:

$$\mathrm{Var}(e_i) = \delta_i^2 = \frac{\sigma^2}{w_i}$$

Where $w_1, \cdots, w_n$ are known positive numbers

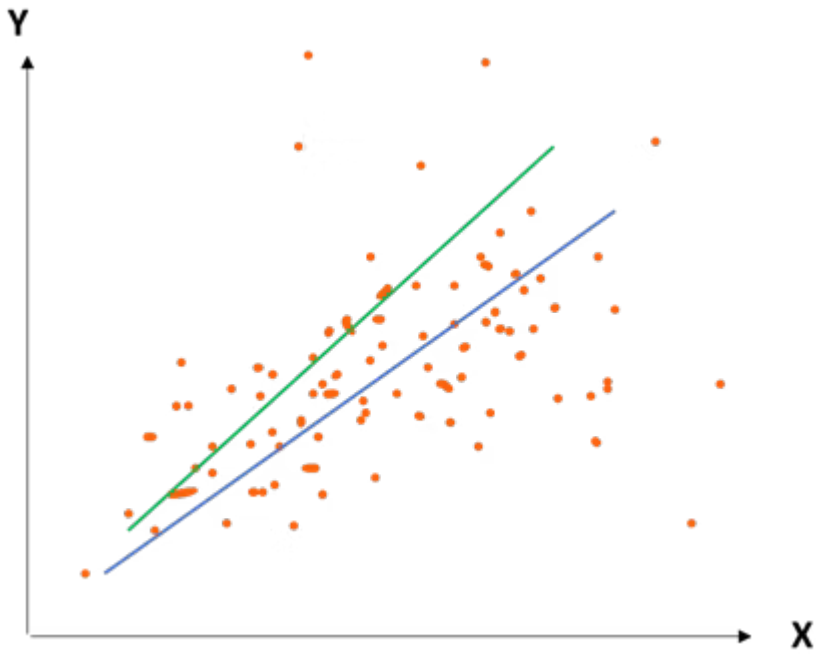
For weighted least square regression, we need to solve for coefficients by minimizing the following sum of weighted squared residuals.

$$RSS(\beta_0, \beta_1) = \sum_{i=1}^{n} w_i (Y_i - \beta_0 - \beta_1 X_i)^2$$

The above function is called an **objective function**. An **objective function** is a function we would like to maximize or minimize. Please see section 4 in this lesson for more information on how to derive the coefficients of weighted least square regression.

We can see the weight w_i is inversely proportional to the variance of each observation. It means for an observation that is highly variable, it will be given a smaller weight for its weighted squared residual. Hence, this observation will have less importance in the minimization process. For an observation that has small variability, its weight will be larger. Its weighted squared residual will be larger, and it will be more important in the minimization process.

Figure 2: Impact of Data Weighting on Regression Lines



From Figure 2, Y is the dependent variable and X is the independent variable of a regression. Figure 2 shows the scatterplot of the two variables. From the graph, we can see the data is not homoscedastic because there is more variation of Y when X increases. If we do not apply weights to the data and run an OLS regression model, then the regression line will try to treat all data points equally. The result line will be like the green line. The green line is higher and steeper towards the data point on the top of the graph because OLS treats every point equally. However, if we apply weights to all the data points, we will reduce the impact of those points that are further away from where most of the points cluster together. The weighted least square (WLS) regression line will be like the blue line. WLS focuses more on the data points that are close to each other and gives lower weight to points that are farther away. Hence, WLS is also a good method to handle **outliers**. When there is an outlier in the dataset, by the method of WLS, it will give less weight so this outlier will have less impact on the result of the WLS regression model.

One thing to notice is that if the weight is 1, then we are basically running an OLS regression model. In the field of regression analysis, OLS is actually a special case of WLS because OLS has weights equal to 1.

2.3 How Do We Estimate Weights?

We don't always know the numbers of the weight. What should we do when the weights are unknown? In the last section, we learned that the weight is proportionally inverse to the variance of each error term.

$$w_i = \frac{\sigma^2}{\sigma_i^2}$$

Since w_i is the weight and σ^2 is an unknown constant and we only care about relationship among weights, not the scale, we can assume σ^2 to be 1. As such, we can try

to estimate the following weight function.

$$w_i = \frac{1}{\sigma_i^2}$$

However, we still don't know the variance for each error term of the observation (σ_i^2). We will need to estimate the variances of error terms.

Here are the steps to run a WLS regression model:

Step 1: Run an OLS regression model.

Step 2: Draw a scatter plot with fitted values from OLS on the X axis and residuals from OLS on the Y axis.

Step 3: Inspect the scatter plot to see if there is a cone or megaphone pattern in the plot to indicate heteroskedasticity. You can also use the **Breusch-Pagan test** to check for heteroskedasticity, as discussed in the last section.

Step 4: If you determine that there is heteroskedasticity in the OLS model, you will have to estimate the weights. We know from the last section that the weight for each observation is the inverse of the variance. We can use squared residuals from OLS to estimate the variances. It also means to use residuals from OLS to estimate the standard deviations. In order to estimate squared residuals, we will run an OLS regression model with absolute values of the residuals as the dependent variable and the fitted values from OLS model in **Step 1** as an independent variable as follows:

$$|\text{residuals from OLS}| = 1 + \gamma \cdot \text{fitted values from OLS} + e$$

We will use the fitted value from the above OLS regression as the proxy for standard deviation. Variance is the squared standard deviation. We then plug the estimated standard deviation into the following equation to get the weight variable.

$$\text{weight} = \frac{1}{\text{standard deviation}^2}$$

Step 5: Once we get the weight variable, we can plug this weight variable into any statistical software to run a weighted least square regression model.

3. Foreign Exchange Market Analysis Using Weighted Least Square Regression

Now, let's look at an example to learn how to run weighted least square regression.

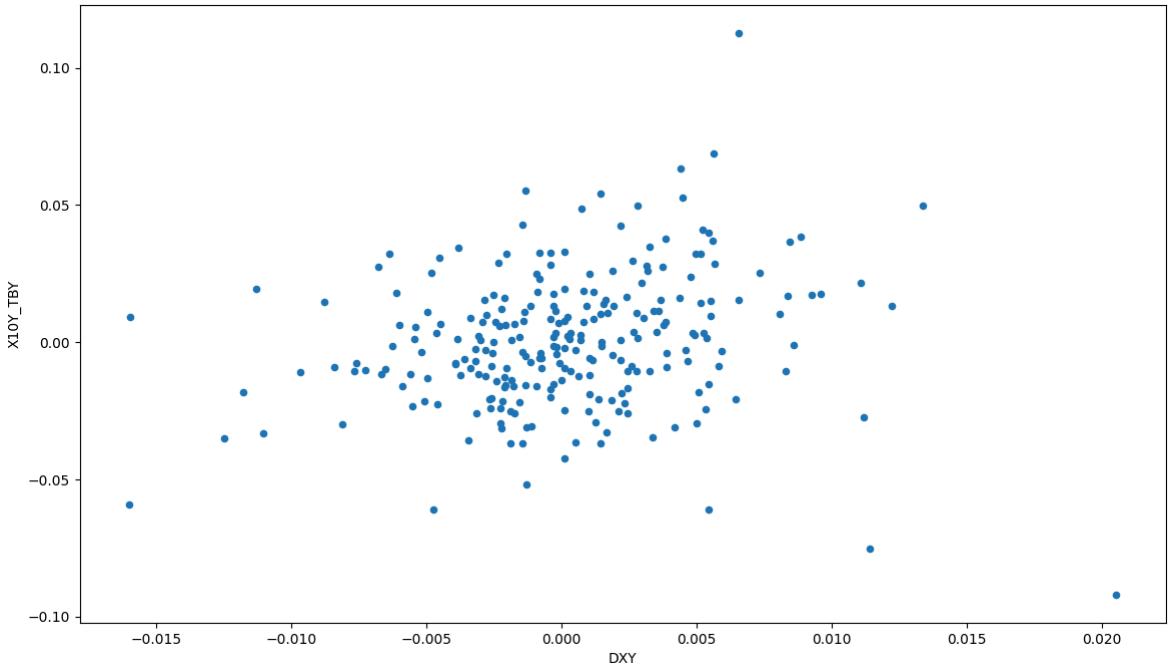
We will use WLS to understand how the U.S. 10-year treasury bond yield would influence the U.S. Dollar Index. Both variables are converted to daily returns for analysis. We use the daily data from 2016. Here are their variable names.

- *DXY*: US Dollar Index daily return
- *X10Y_TBY*: 10-Year Treasury Bond Yield daily return

First, let's draw a scatter plot to see if the two variables have any relationship.

Figure 3: U.S. Dollar Index Daily Return and 10-Year Treasury Bond Yield Daily Return Scatter Plot

In [3]:
```
# Create scatter plot
scatt = data_set.plot.scatter(x="DXY", y="X10Y_TBY")
scatt.set(xlabel="DXY", ylabel="X10Y_TBY")
plt.show()
```



From Figure 3, we can see two interesting patterns. First, they seem to have a positive relationship. We can see as *DXY* increases, *X10Y_TBY* increases too. Secondly, as *DXY* increases, the data points spread out more. It forms a cone or megaphone shape. It indicates that there might be a heteroskedasticity issue.

Now, let's run an OLS regression with *DXY* as the dependent variable and *X10Y_TBY* as the independent variable.

Figure 4: OLS Regression Result

In [4]:
```
# OLS model
model_1 = smf.ols("DXY ~ X10Y_TBY", data=data_set).fit()
model_1.summary()
```

Out[4]:

OLS Regression Results

Dep. Variable:	DXY	**R-squared:**	0.030
Model:	OLS	**Adj. R-squared:**	0.026
Method:	Least Squares	**F-statistic:**	7.738
Date:	Fri, 05 Dec 2025	**Prob (F-statistic):**	0.00582
Time:	18:18:12	**Log-Likelihood:**	986.78
No. Observations:	250	**AIC:**	-1970.
Df Residuals:	248	**BIC:**	-1963.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P>|t|**	**[0.025**	**0.975]**
Intercept	0.0001	0.000	0.474	0.636	-0.000	0.001
X10Y_TBY	0.0336	0.012	2.782	0.006	0.010	0.057

Omnibus:	26.741	**Durbin-Watson:**	1.909
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	94.918
Skew:	0.322	**Prob(JB):**	2.45e-21
Kurtosis:	5.949	**Cond. No.**	40.7

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

In [5]:
```
# Parameters with high precision
print(model_1.summary2().tables[1])
```

```
              Coef.  Std.Err.         t     P>|t|    [0.025    0.975]
Intercept  0.000141  0.000297  0.474101  0.635845 -0.000444  0.000725
X10Y_TBY   0.033568  0.012067  2.781725  0.005823  0.009800  0.057336
```

From Figure 4, we can see that *DXY* and *X10Y_TBY* have a positive coefficient, which is consistent with what we saw in the scatter plot. The coefficient is also significant. The R^2 is not very good for this simple regression model. Now, let's check the scatter plot for residuals and fitted values of this OLS model to see if there is a heteroskedasticity issue.

Figure 5: Scatter Plot for OLS Fitted Values and OLS Residuals

In [6]:
```
# Scatter Plot for OLS Fitted Values and OLS Residuals

# Fitted values and residuals
model_fitted_y = model_1.fittedvalues
model_resid_y = model_1.resid

# Plot
```

```
plt.scatter(x=model_fitted_y, y=model_resid_y)
plt.title("OLS Residuals vs OLS Fitted Values")
plt.xlabel("OLS Fitted values")
plt.ylabel("OLS Residuals")
plt.show()
```

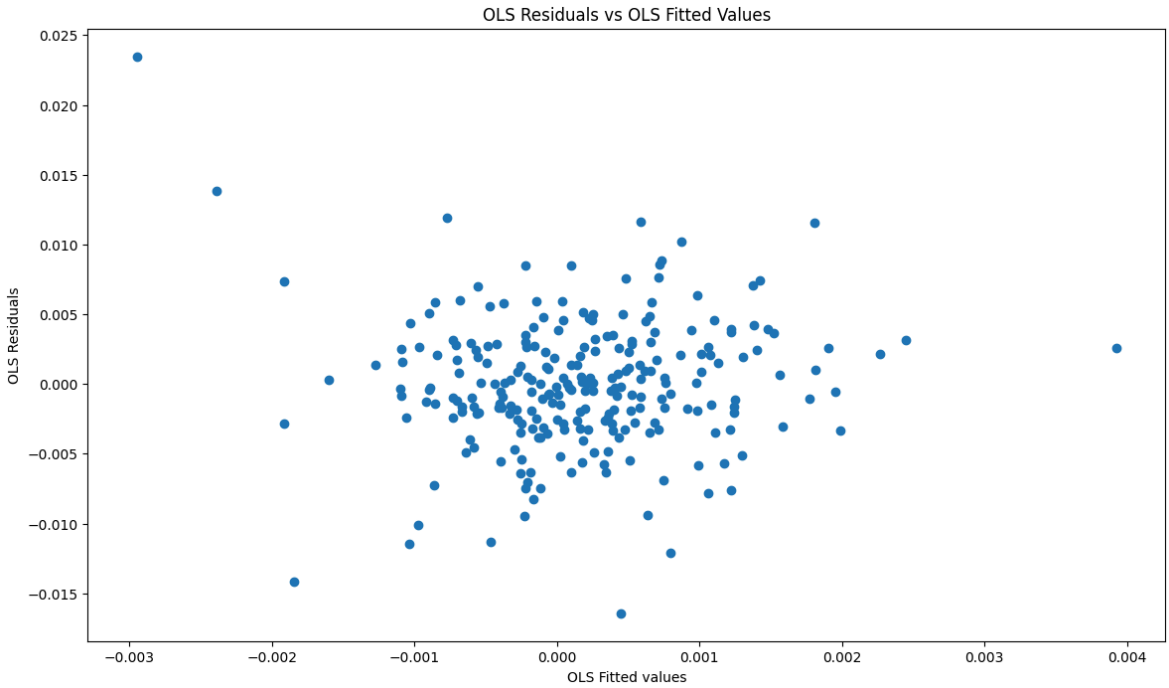


Figure 5 shows that there is more variation in residuals when fitted values are negative than when fitted values are positive, especially when fitted values are below -0.002. The scatter plot indicates there might be a heteroskedasticity issue. Let's run a Breusch-Pagan test to double-check the hypothesis.

Figure 6: Breusch-Pagan Test Result

In [7]:
```
# Breusch-Pagan Test
name = ["Lagrange multiplier statistic", "p-value", "f-value", "f p-value"]
test = sms.het_breuschpagan(model_1.resid, model_1.model.exog)
pd.DataFrame(test, index=name, columns=[""])
```

Out[7]:

Lagrange multiplier statistic	13.897238
p-value	0.000193
f-value	14.597521
f p-value	0.000168

The test result from Figure 6 also confirms the existence of heteroskedasticity because the p-value is less than 0.05. Based on the above information, we will run the model with weighted least square, with the weight generated from the formula from the last section.

Figure 7: WLS Result

In [8]:
```
# WLS regression result

# Add Absolute residuals and fitted values to dataset columns
```

```
data_set["abs_residuals"] = np.abs(model_1.resid)
data_set["fitted_values"] = model_1.fittedvalues

# Fit OLS model with absolute residuals and fitted values
model_temp = smf.ols("abs_residuals ~ fitted_values", data=data_set).fit()

# Compute weights and add it to the data_set column
weights = model_temp.fittedvalues
weights = weights**-2
data_set["weights"] = weights

# Fit WLS model
Y = data_set["DXY"].tolist()
X = data_set["X10Y_TBY"].tolist()
X = sm.add_constant(X)  # add a intercept point

model_WLS = sm.WLS(Y, X, data_set["weights"]).fit()
model_WLS.summary()
```

Out[8]:

WLS Regression Results

Dep. Variable:	y	**R-squared:**	0.100
Model:	WLS	**Adj. R-squared:**	0.096
Method:	Least Squares	**F-statistic:**	27.59
Date:	Fri, 05 Dec 2025	**Prob (F-statistic):**	3.23e-07
Time:	18:18:12	**Log-Likelihood:**	994.52
No. Observations:	250	**AIC:**	-1985.
Df Residuals:	248	**BIC:**	-1978.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	0.0001	0.000	0.346	0.730	-0.000	0.001
x1	0.0516	0.010	5.253	0.000	0.032	0.071

Omnibus:	10.978	**Durbin-Watson:**	1.965
Prob(Omnibus):	0.004	**Jarque-Bera (JB):**	23.475
Skew:	0.063	**Prob(JB):**	7.99e-06
Kurtosis:	4.496	**Cond. No.**	35.0

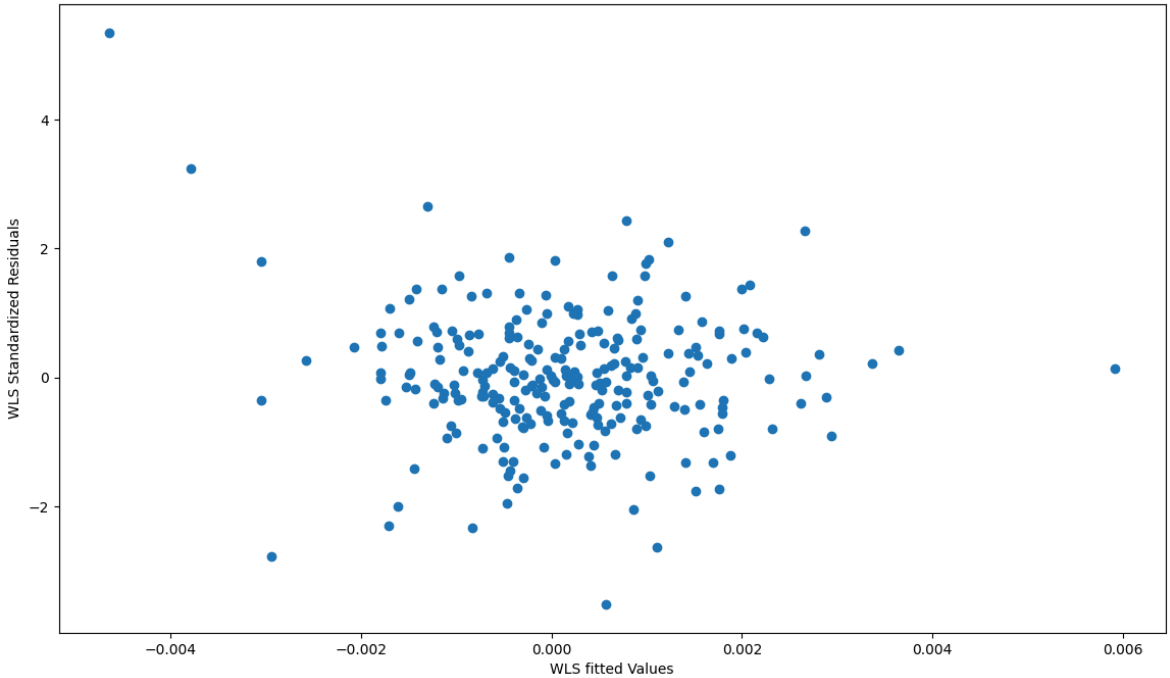
Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

From Figure 7, we can see *X10Y_TBY* still has a positive coefficient and the coefficient is still significant. The R^2 also improves a lot from 0.03 to 0.10. Now, let's look at the residuals and fitted values from WLS to see if we can correct the heteroskedasticity issue.

Figure 8: WLS Standardized Residuals and Fitted Values Scatter Plot

In [9]:
```
# WLS fitted values and standardized residuals
fitted_valuesWLS = model_WLS.fittedvalues
model_residWLS = model_WLS.resid
std_model_residWLS = model_residWLS / model_residWLS.std(ddof=1)

# Plot WLS fitted v. vs standardized res. scatterplot
plt.scatter(fitted_valuesWLS, std_model_residWLS)
plt.xlabel("WLS fitted Values")
plt.ylabel("WLS Standardized Residuals")
plt.show()
```



4. Derivation of Coefficients for Weighted Least Square Regression

We first use a single regression model to derive coefficients for weighted least square regression.

Assume we have the following simple regression model for n observations:

$$Y_i = \beta_0 + \beta_1 X_i + e_i, \quad i = 1, \ldots, n$$

Where $e \sim N(0, \frac{\sigma^2}{w_i})$ and $w_1, \ldots, w_n$ are known positive numbers

For weighted least square regression, we are looking for β_0 and β_1 that will minimize the sum of weighted squared residuals. Here is the objective function:

$$RSS(\beta_0, \beta_1) = \sum_{i=1}^{n} w_i (Y_i - \beta_0 - \beta_1 X_i)^2$$

We take partial derivatives of the objective function with respect to β_0 and β_1 to get the WLS estimates of $\hat{\beta}_0$ and $\hat{\beta}_1$

$$\hat{\beta}_0 = \overline{Y_w} - \hat{\beta}_1 \overline{X_w}$$
$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} w_i (X_i - \overline{X_w})(Y_i - \overline{Y_w})}{\sum_{i=1}^{n} w_i (X_i - \overline{X_w})^2}$$

Where $\overline{X_w}$ and $\overline{Y_w}$ are weighted averages of X and Y with weights w.

$$\overline{X_w} = \frac{\sum_{i=1}^{n} X_i w_i}{\sum_{i=1}^{n} w_i} \quad \text{and} \quad \overline{Y_w} = \frac{\sum_{i=1}^{n} Y_i w_i}{\sum_{i=1}^{n} w_i}$$

We can apply the same process above to solve for coefficients for weighted least square regression with multiple independent variables. This is a general form of derivation because we don't restrict the number of independent variables. Since we have multiple independent variables now, we will use the matrix form to solve the minimization problem.

For multiple weighted least regression, we have the model:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{e}$$
$$\text{Var}(e) = \sigma^2 \mathbf{W}^{-1}$$

We minimize the following sum of weighted squared residuals in matrix form for multiple weighted least square regression models. Here is the objective function:

$$RSS(\boldsymbol{\beta}) = \sum_{i=1}^{n} w_i (Y_i - X_i^t \boldsymbol{\beta})^2 = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^t \mathbf{W} (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})$$

We take the partial derivatives of the above objective function with respect to $\boldsymbol{\beta}$ and set the equations to 0 to solve for $\boldsymbol{\beta}$.

$$\sum_{i=1}^{n} w_i e_i X_i^t = 0$$

The general form of WLS estimator is given by

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^t \mathbf{W} \mathbf{X})^{-1} (\mathbf{X}^t \mathbf{W} \mathbf{Y})$$

5. Conclusion

In this lesson, we first reviewed the key assumptions for an ordinary least square regression model. We then relaxed the assumption of homoscedasticity. If the data shows heteroskedasticity, which is a violation of homoscedasticity, we will use a weighted least square regression model to handle the issue.

We pointed out that OLS is a special case for WLS when the weight is 1, which means the data is unweighted. We then introduced the basics of WLS and the steps to run WLS. We finished the lesson with one example from the foreign exchange market. In the next

lesson, we will introduce another regression method when the assumptions of OLS do not hold up well in the dataset.

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